



Statistical Methods & Machine Learning 1

Class #2

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Outline

- Population, sample, and sampling distribution
- Normal distribution and central limit theorem
- Hypothesis testing
- Assessing normality
- Lab

Population vs. sample

- Use statistics to make some inferences or predictions about a particular population of interest.
- However, we will rarely have data on the entire population, so what do we do?
 - We can take a sample.
- A sample is subset of your population of interest that data is gathered on.
- If done well, statistical analyses on this sample can be generalized to the population of interest with some uncertainty.
 - Probability distributions allow us to do this while quantifying this uncertainty when certain conditions are met.

Population vs. sample

- Can differentiate by parameter notation:
 - Population parameters: Y, β, μ, σ
 - Sample parameters: $\hat{Y}, \hat{\beta}, \bar{x}, s$
- Most of the time we are working with a sample rather than full population.

Degrees of freedom

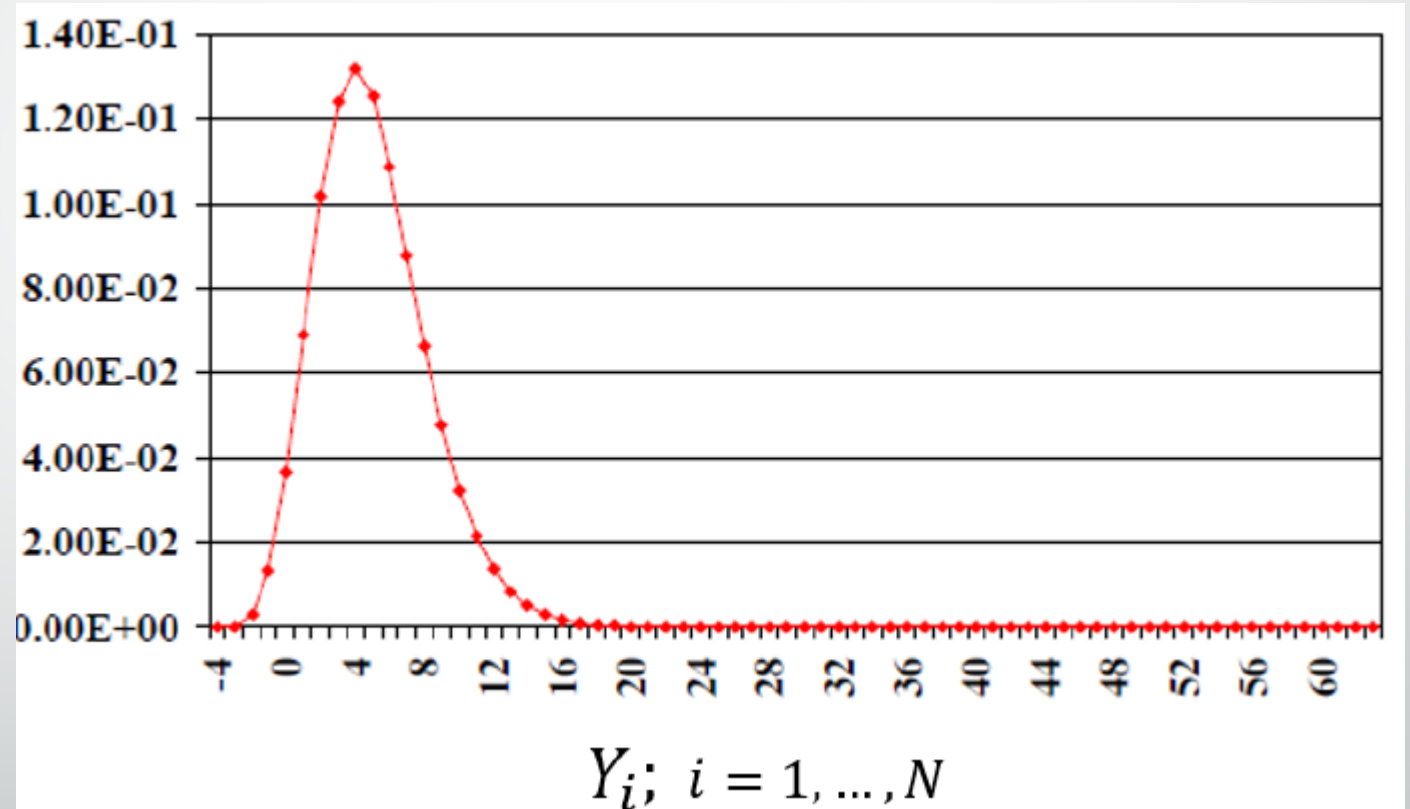
- When you answer the question "How many data points are required to estimate a parameter of interest (e.g., $\bar{Y}$)?" your degrees of freedom are what's left.
- Can calculate this using the formula $n-p$ (sample size - # of parameters).
- In regression models this formula is often written as $n-p-1$ to distinguish between the intercept and other parameters.

Normal Distribution

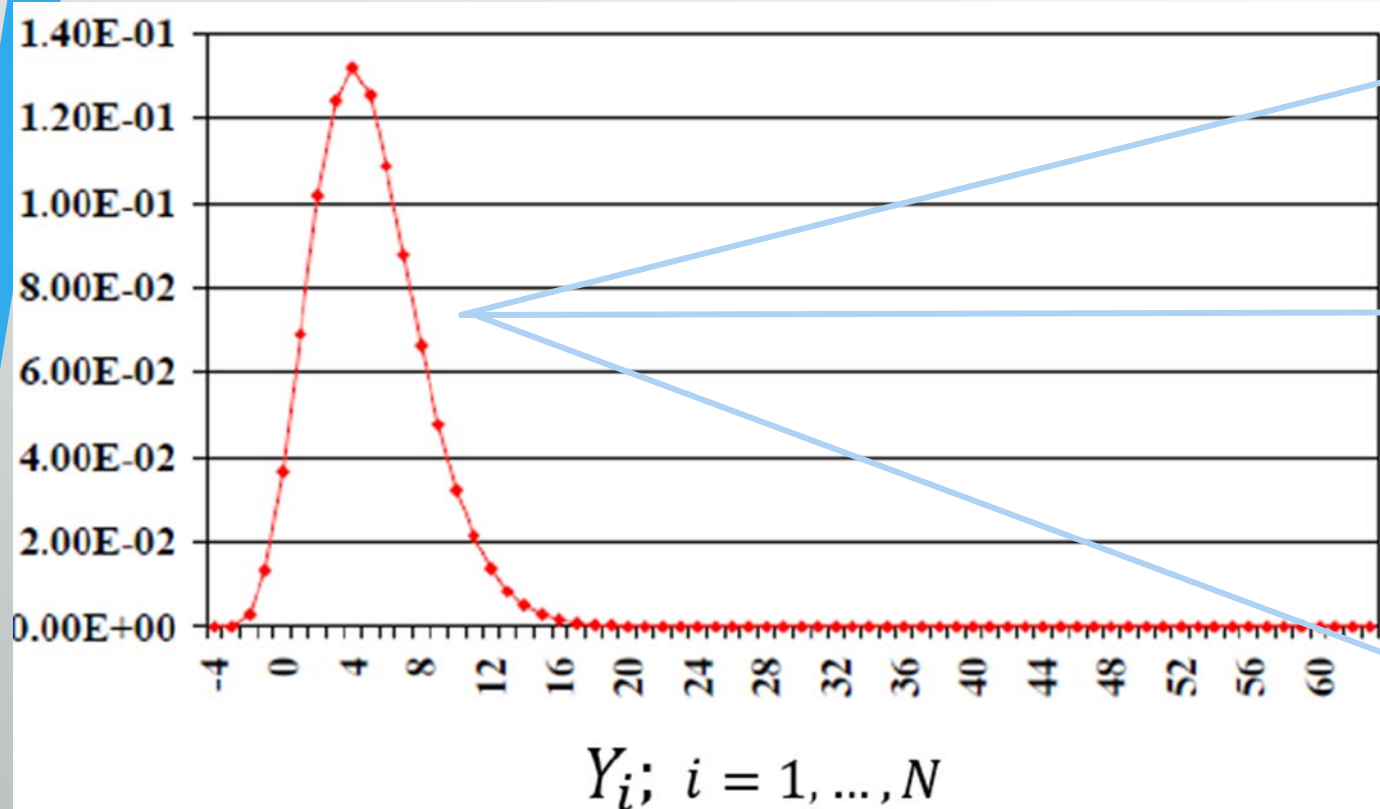
- Often assumed for continuous variables
- Reasons:
 - Bell-curve shape
 - Central Limit Theorem (CLT)
 - Variables can be transformed to normal
 - Well-developed theories

Population Distribution: Probability Distribution of Y_i

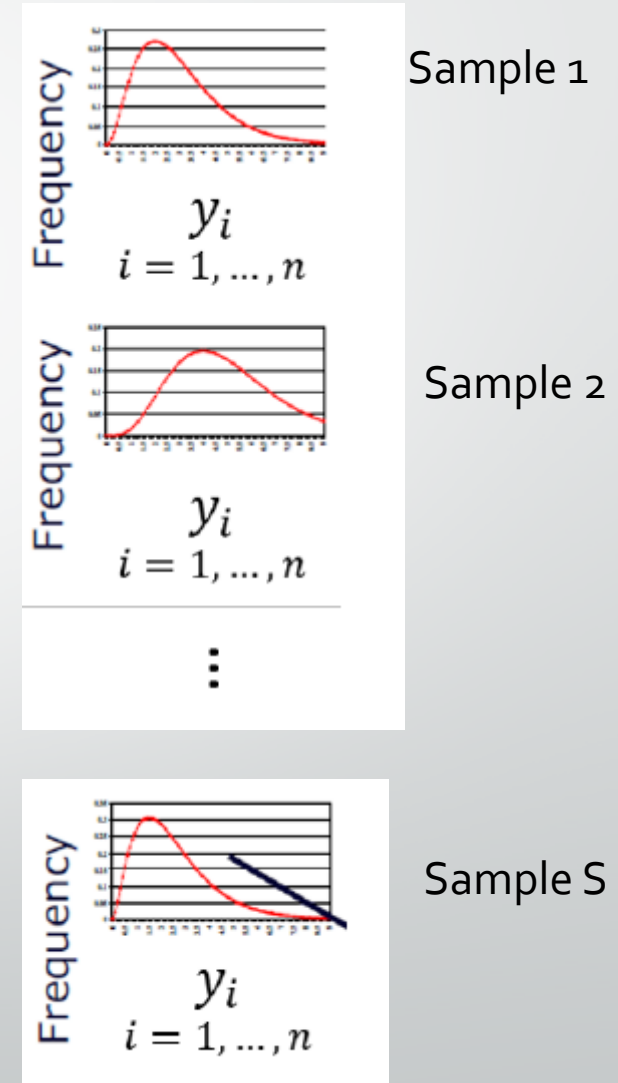
- What are Y_i ?
- How are Y_i in the population distributed?



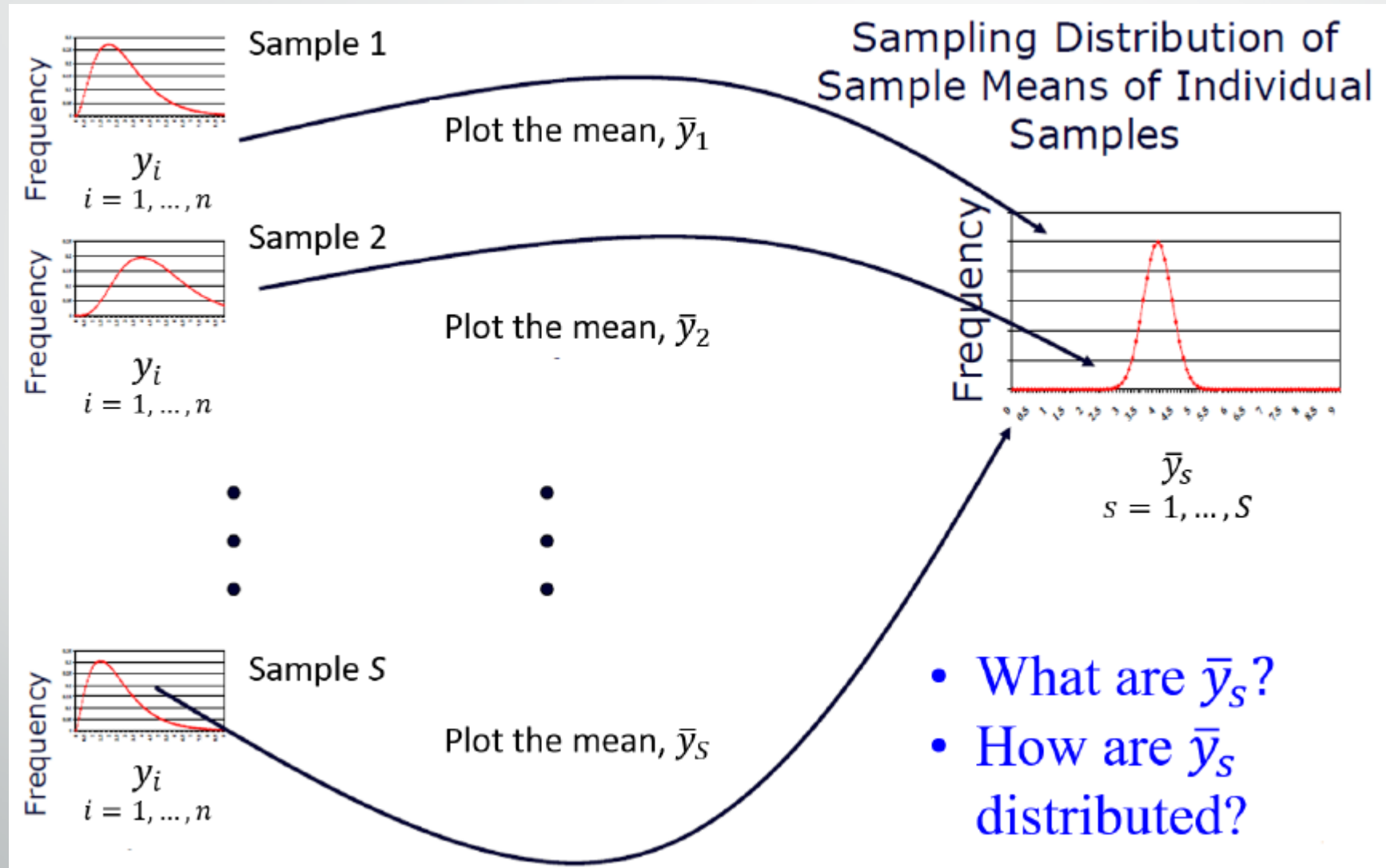
Sample Distribution: Probability Distribution of y_i



- What are y_i ?
- How are y_i in each sample distributed?

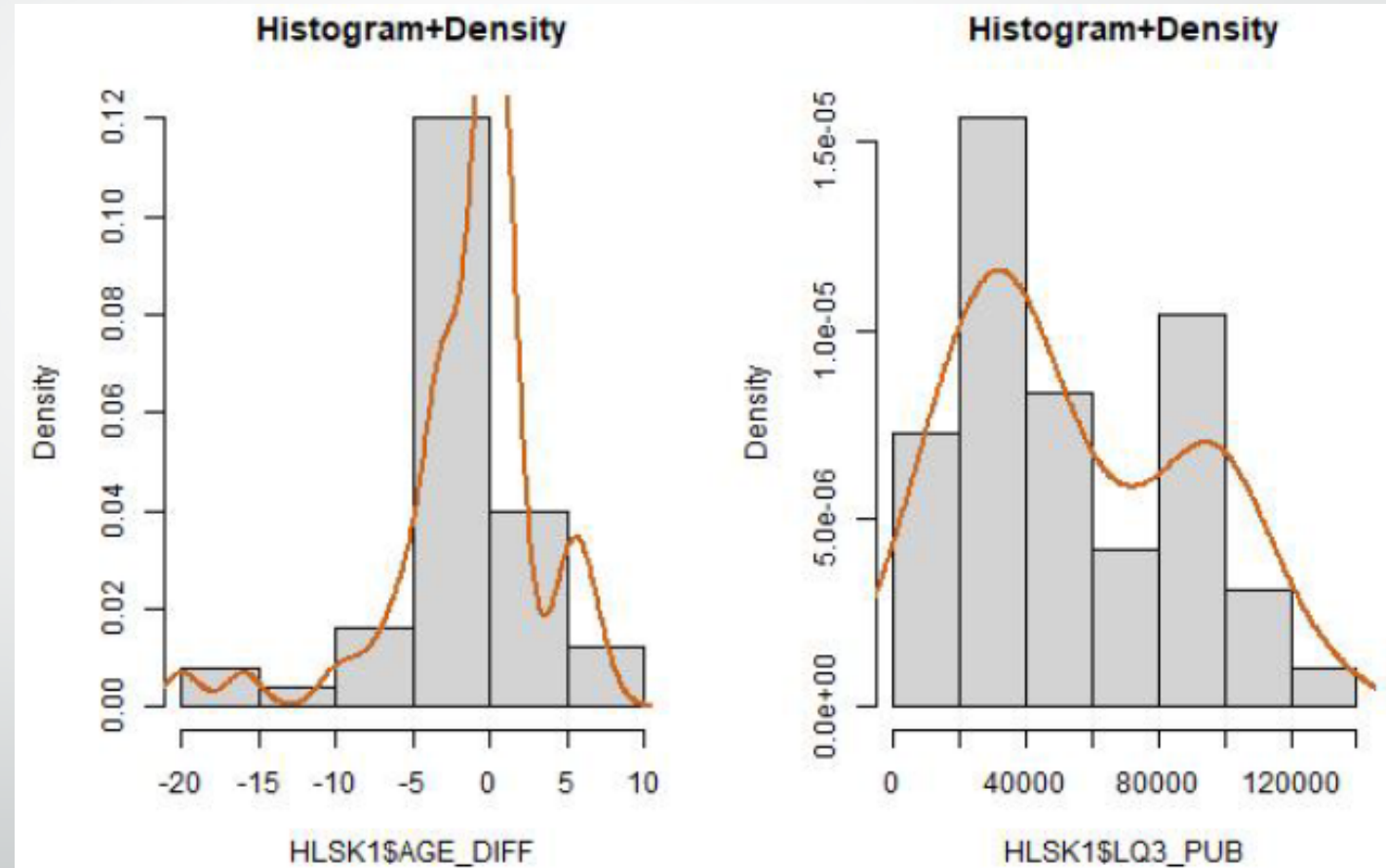


Sampling Distribution: Probability Distribution of $\bar{y}_s$



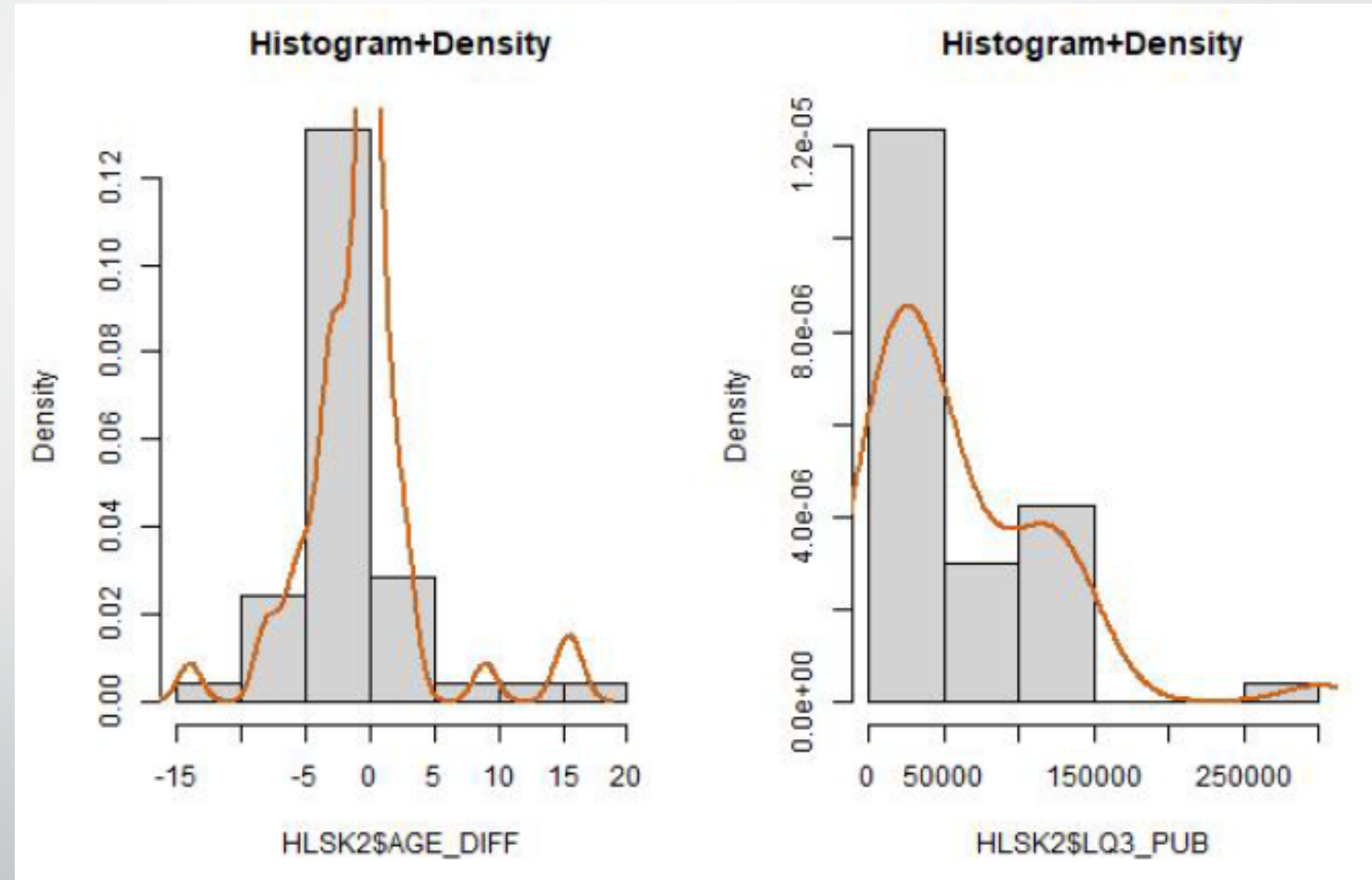
Example: Sampling Distribution - 1

- Now assume HLSK data is from an entire population. Draw a simple random sample of $n = 50$ from HLSK data; distributions of AGE_DIFF and LQ3_PUB appear as:



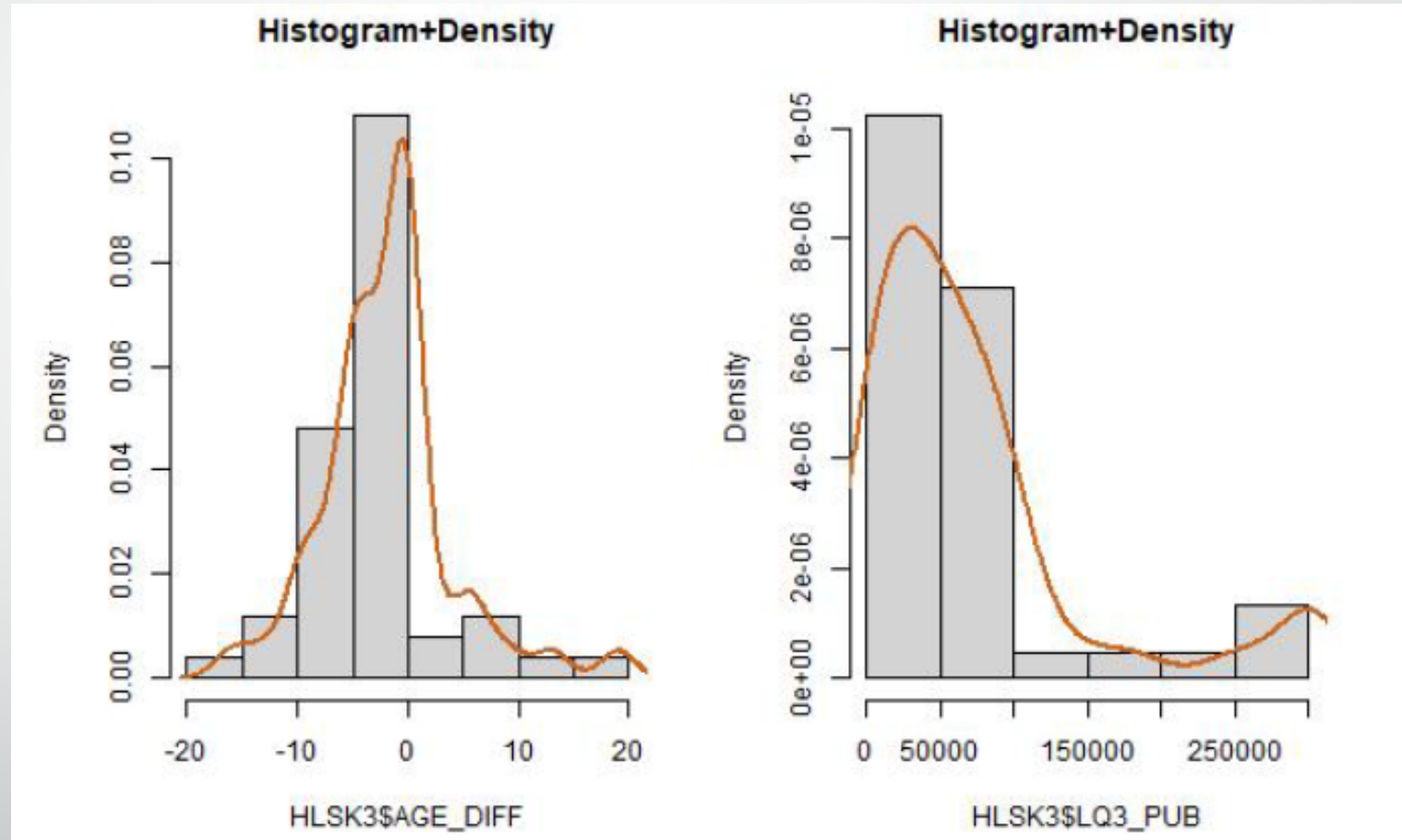
Example: Sampling Distribution - 2

- Draw another simple random sample of $n = 50$ from the HLSK data; distributions of AGE_DIFF and LQ3_PUB look like:



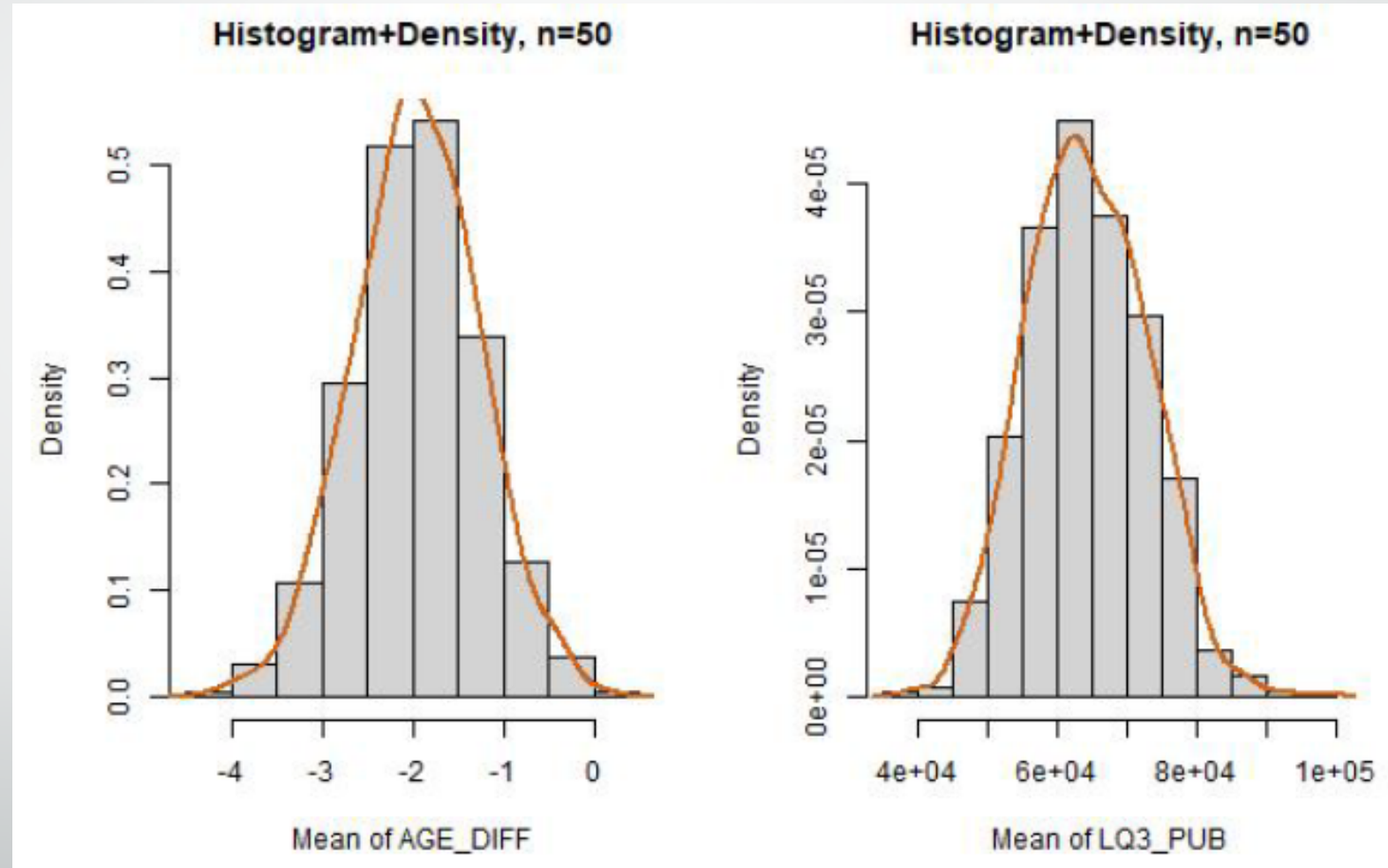
Example: Sampling Distribution - 3

- Draw another simple random sample of $n = 50$ from the HLSK data; distributions of AGE_DIFF and LQ3_PUB look like:



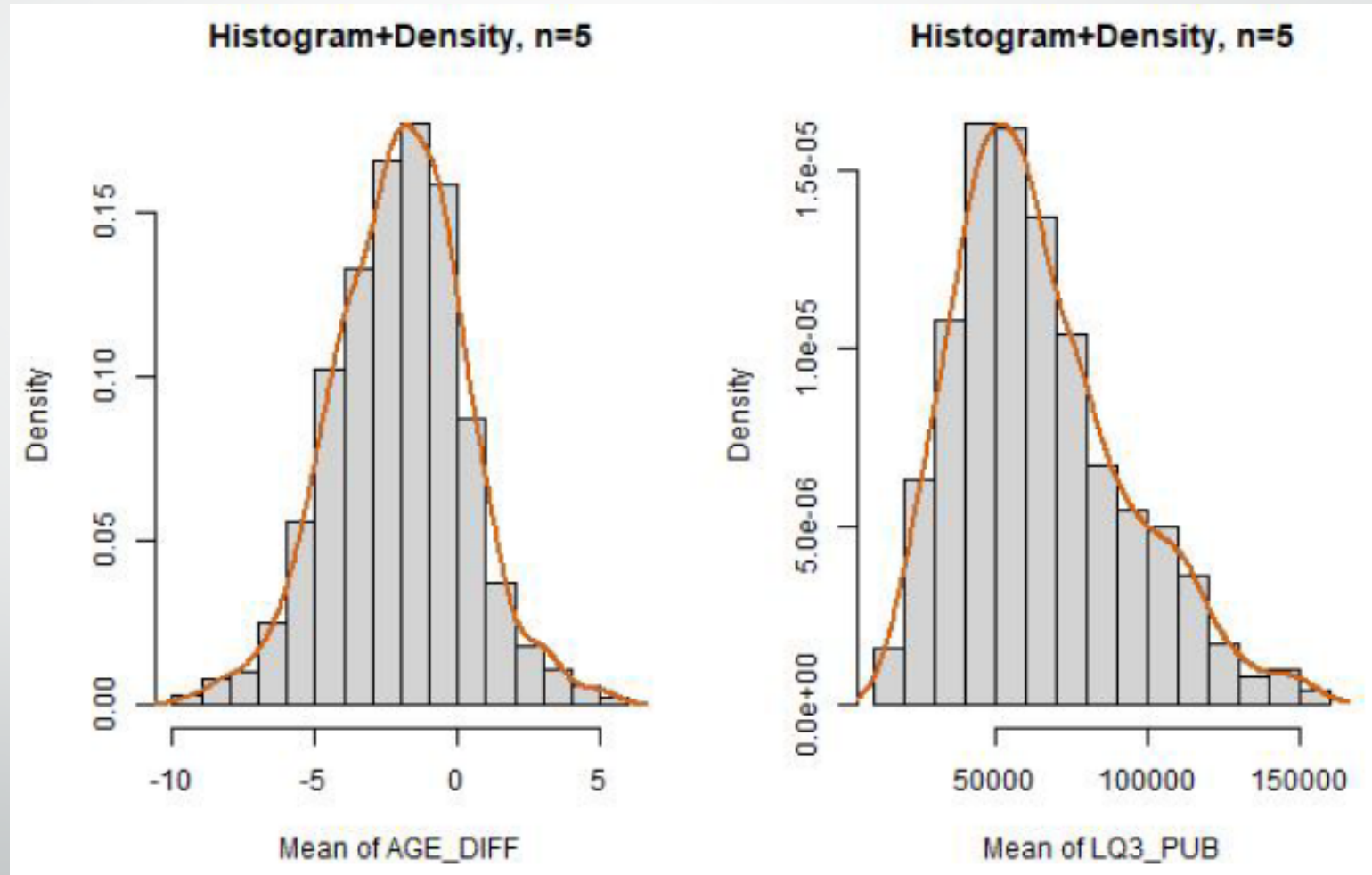
Example: Sampling Distribution - 4

- Repeat this 1000 times; take the mean of each sample and plot the means, $\bar{y}_S$, of all 1000 samples;
- Distributions of the sample means look like:



Example: Sampling Distribution - 5

- What about when $n = 5$?
- Repeat this 1000 times; take the mean of each sample and plot the means, $\bar{y}_S$, of all 1000 samples
- Distributions of the sample means look like:



Probability vs. Sampling Distribution

- A random variable, y (e.g., AGE_DIFF or LQ3_PUB), is subject to a probability distribution with expectation (μ) and variance (σ^2)
 - E.g., If the shape of the distribution of a continuous random variable, y , is normal then $y \sim N(\mu, \sigma^2)$
- The mean estimated from a sample, $\bar{y}_S$, is also a random variable subject to a probability distribution with its own expectation and variance due to variability in sampling
- Probability distribution of $\bar{y}_S$ is **sampling distribution**
- When n is sufficiently large, $\bar{y} \sim N(\mu, \frac{\sigma^2}{n})$ due to the **Central Limit Theorem**, even if y does not follow a normal distribution



Hypothesis testing

Hypothesis testing

- Commonly want to make comparisons involving groups and/or quantities
- Can be comparison between multiple groups
 - Is the mean height of students in class 1 different from that in class 2?
- Can compare against a specified quantity
 - Is mean change different from 0?
- These are typically estimates, so there is some uncertainty
 - How do we decide if an estimate is likely to be different given this uncertainty?
 - **Hypothesis testing**

Hypothesis testing

- Null and alternative hypotheses
 - Two mutually exclusive hypotheses about a population, relationship, or quantity.
- A hypothesis test allows us to evaluate which hypothesis the data best support.
- Null hypothesis (H_0): there is no difference or effect in the population.
- Alternative hypothesis (H_A): there is a difference or effect in the population.
- If there is sufficient evidence supporting H_A we reject the null hypothesis.
 - If there is not sufficient evidence supporting H_A we **fail to reject the null hypothesis**.
 - How do we determine there is *sufficient* evidence?

Hypothesis testing

Can conduct one-sided or two-sided test

- One-sided means that we are testing whether the mean is either greater than or less than another quantity.
 - Less common and should always be supported by prior knowledge.
 - $H_0: \mu_1 \geq \mu_2; H_A: \mu_1 < \mu_2$
 - $H_0: \mu_1 \leq \mu_2; H_A: \mu_1 > \mu_2$
- Two-sided means that we are testing whether the mean is simply not equal to another quantity.
 - More conservative, generally better practice unless prior evidence, and/or specific question where two-sided doesn't make sense.
 - $H_0: \mu_1 = \mu_2; H_A: \mu_1 \neq \mu_2$

Hypothesis testing

Setting alpha, α , and calculating p-values

- α is a standard set by researcher that states how strongly the data must contradict the null hypothesis for it to be rejected.
- $\alpha = 0.05$ is most common, but **this is merely convention**
 - It's saying we are ok with a 5% chance of identifying an effect (rejecting the null hypothesis) when, in reality, there is no effect.
- We calculate a p-value to assess whether this pre-specified threshold has been met.
 - Specifically, it's the probability of seeing a value as or more extreme as what we see in the data.

P-values and their interpretation

- With $\alpha = 0.05$ p-values < 0.05 are often called "significant" – **use this approach with caution**
- How do we interpret p-values correctly?
- If we were to have $p = 0.01$ we would say that **assuming the null hypothesis is true** there is a 1% chance of seeing a value as or more extreme than the value (e.g., mean) from the data.
- It boils down to the probability that we **incorrectly** reject the null hypothesis if the null hypothesis is true.

Confidence intervals

- Related to p-values, but a bit more informative.
- Convention is to use 95% confidence intervals which corresponds to $\alpha = 0.05$, but again, this is arbitrary.
- Follow the general formula of: $\bar{y} - z_{\frac{\alpha}{2}}(\sqrt{\frac{\sigma^2}{n}}), \bar{y} + z_{\frac{\alpha}{2}}(\sqrt{\frac{\sigma^2}{n}})$
- Will be small tweaks to this formula depending on the situation as we will see.

Confidence intervals

- How to interpret 95% CI
- Ex. You estimate the mean wage (per hour) of a sample of employees to be \$25 (95% CI: 20, 30).
- How would you interpret this?

Confidence intervals

- How to interpret 95% CI
- Ex. You estimate the mean wage (per hour) of a sample of employees to be \$25 (95% CI: 20, 30). You would interpret the 95% CI to mean that if you were to take repeated samples of the employees (calculating the sample mean and 95% CI each time) the **true mean wage would be contained in the confidence interval 95% of the time.**

Hypothesis tests for different scenarios

- Single sample - Known σ^2
- Single sample - Unknown σ^2
- Two independent samples - Unknown and equal σ^2
- Two independent samples - Unknown and unequal σ^2
- Dependent/paired samples - Unknown, unequal σ^2

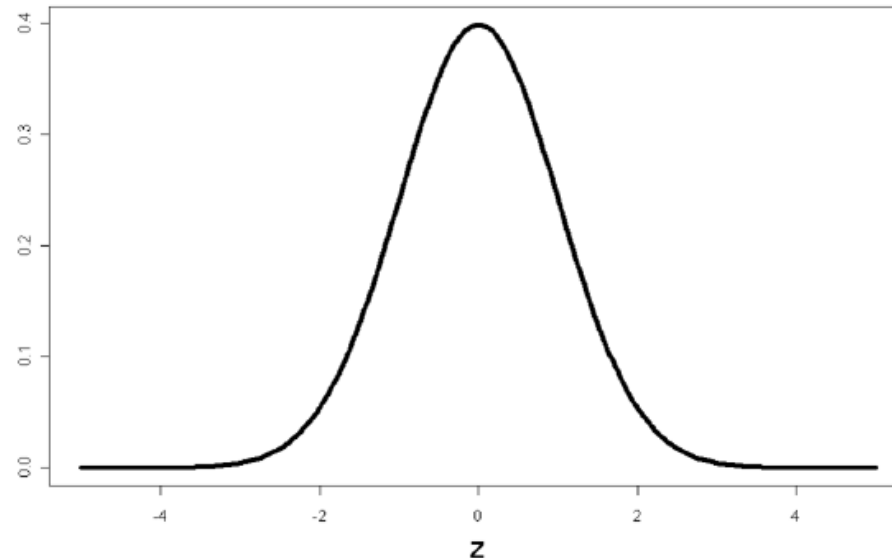


Single Sample – Known σ^2

Single sample known σ^2

- When we have a continuous variable, Y , where we know σ^2 and we want to compare the mean of Y with specific quantity.
- Can use a one-sample Z test

Type	H_0	H_A	Rejection Region	Observed p -value
Two-sided	$\mu = \mu_0$	$\mu \neq \mu_0$	$ z_{obs} > z_{\alpha/2}$	$2P(Z > z_{obs})$
One-sided	$\mu \leq \mu_0$	$\mu > \mu_0$	$z_{obs} > z_{\alpha}$	$P(Z > z_{obs})$
	$\mu \geq \mu_0$	$\mu < \mu_0$	$z_{obs} < -z_{\alpha}$	$P(Z < z_{obs})$





Single Sample – Unknown σ^2

Hypothesis Testing and Confidence Interval for μ

Since σ^2 is not known, use estimated variance, $s^2 = \frac{\sum_{i=1}^n (y_i - \hat{y})^2}{n-1}$, from a sample instead. Leads to standardized version:

$$t_{obs} = \frac{\hat{y} - \mu}{\sqrt{s^2 n^{-1}}}$$

which follows a Student's t-distribution ($\approx$ normal dist.)

Hypothesis testing and confidence interval calculation follow those for variables from normal distribution with known σ^2 .

Same procedures as with z test, but substitute z_{obs} with t_{obs} and $z_{\alpha/2}$ with $t_{(\alpha/2, n-1)}$

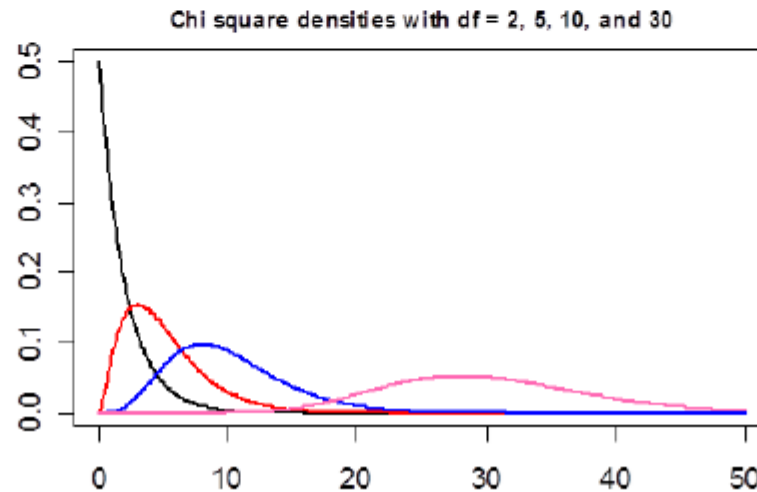
Sample variance, s^2

s^2 is estimated and subject to variance and has its own distribution with

$$E(s^2) = \sigma^2 \text{ and } V(s^2) = \frac{2\sigma^4}{n-1}, \text{ estimated by } \hat{V}(s^2) = \frac{2s^4}{n-1}$$

Under normality, $(n-1)\frac{s^2}{\sigma^2} \sim \chi_{n-1}^2$

$$\Rightarrow 100 \times (1 - \alpha)\% \text{ CI for } \sigma^2 : \left[\frac{(n-1)s^2}{\chi_{(n-1), \frac{\alpha}{2}}^2}, \frac{(n-1)s^2}{\chi_{(n-1), 1-\frac{\alpha}{2}}^2} \right]$$



Single sample – Unknown σ^2 example

- Class A consists of 35 students with an average height 60 inches. You want know whether this differs from the national average for this age group of 55 inches.
- Can use a one-sample t-test to answer this question

```
1 dat <- rnorm(35, 60, 1)
2 t.test(dat, mu = 55)
```

One Sample t-test

```
data: dat
t = 36.613, df = 34, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 55
95 percent confidence interval:
 59.92068 60.49904
sample estimates:
mean of x
60.20986
```




Two Independent Samples – Unknown, Equal Variance

Testing Differences with Unknown, Equal Variance - 1

$$s_{pooled}^2 = \frac{(n_x - 1)s_x^2 + (n_y - 1)s_y^2}{n_x + n_y - 2}$$

Under normality, $(n_x + n_y - 2) \frac{s_{pooled}^2}{\sigma^2} \sim \chi_{n_x + n_y - 2}^2$

$$V(s_{pooled}^2) = \frac{2\sigma^4}{n_x + n_y - 2}$$

Pooled t -statistic:

$$t_{pooled} = \frac{\bar{x} - \bar{y} - (\mu_x - \mu_y)}{\sqrt{\left(\frac{1}{n_x} + \frac{1}{n_y}\right) s_{pooled}^2}} \sim t_{n_x + n_y - 2}$$

Testing Differences with Unknown Equal Variance - 2

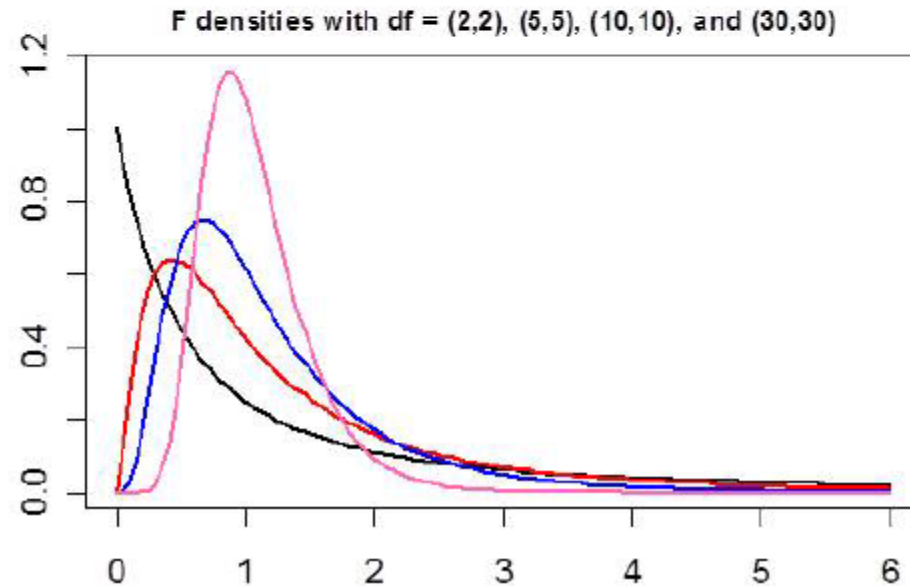
- How do we check if variances are equal?

Testing Equal Variance with F test

Hypothesis: $H_0 : \sigma_x^2 = \sigma_y^2$ vs. $H_A : \sigma_x^2 \neq \sigma_y^2$

$$(n_x - 1) \frac{s_x^2}{\sigma_x^2} \sim \chi_{n_x-1}^2 \text{ and } (n_y - 1) \frac{s_y^2}{\sigma_y^2} \sim \chi_{n_y-1}^2$$

Test F -statistic: $F = \frac{s_x^2}{s_y^2} \sim F_{n_x-1, n_y-1}(\alpha/2)$



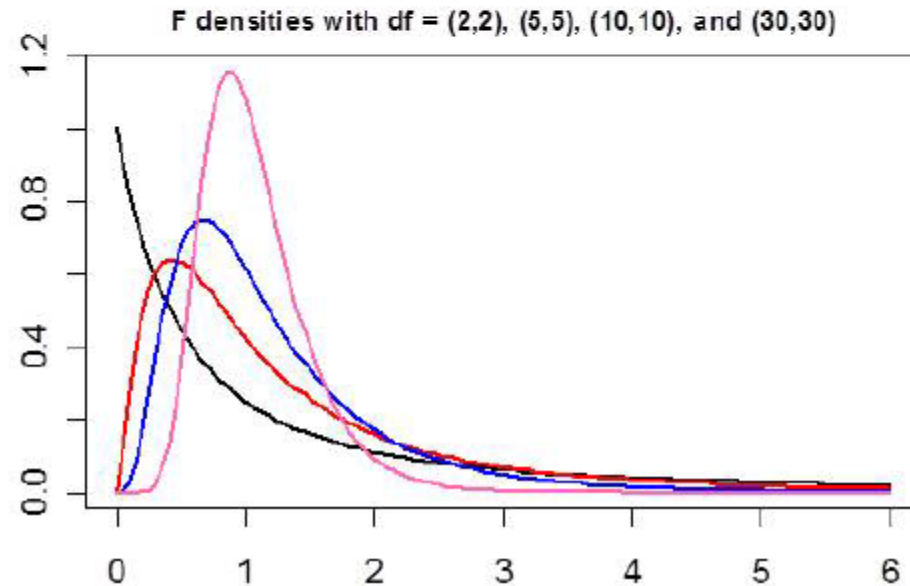
Testing Equal Variance with F test

Hypothesis: $H_0 : \sigma_x^2 = \sigma_y^2$ vs. $H_A : \sigma_x^2 \neq \sigma_y^2$

$$(n_x - 1) \frac{s_x^2}{\sigma_x^2} \sim \chi_{n_x-1}^2 \text{ and } (n_y - 1) \frac{s_y^2}{\sigma_y^2} \sim \chi_{n_y-1}^2$$

Test F -statistic: $F = \frac{s_x^2}{s_y^2} \sim F_{n_x-1, n_y-1}(\alpha/2)$

If we fail to reject null hypothesis can use pooled variance.



Testing Equal Variance with F test

- Want to compare mean birth weight (in pounds) of full-term babies born in two hospitals.
- Based on the results below, what do you conclude?

```
1 hosp1 <- rnorm(200, 7, 2)
2 hosp2 <- rnorm(300, 6.5, 2)
3
4 # Test null hypothesis that variances are equal
5 var.test(hosp1, hosp2, alternative = "two.sided")
```

F test to compare two variances

```
data: hosp1 and hosp2
F = 1.1117, num df = 199, denom df = 299, p-value = 0.4071
alternative hypothesis: true ratio of variances is not
equal to 1
95 percent confidence interval:
 0.8650091 1.4382009
sample estimates:
ratio of variances
 1.111714
```

Testing Equal Variance with F test

- So how do we conduct the appropriate t-test in R?
- What would you conclude based on the results below?

```
1 # Use pooled t-test to compare means
2 t.test(hosp1, hosp2, var.equal=T)
```

Two Sample t-test

```
data: hosp1 and hosp2
t = 1.1717, df = 498, p-value = 0.2419
alternative hypothesis: true difference in means is not
equal to 0
95 percent confidence interval:
 -0.1439741  0.5694074
sample estimates:
mean of x mean of y
 6.824642  6.611925
```



Two Independent Samples – Unknown, Unequal Variance

Testing Differences with Unknown, Unequal Variance

- What do we do if the variances are not equal?

Testing Differences with Unknown, Unequal Variance

- What do we do if the variances are not equal?

```
1 hosp1 <- rnorm(200, 7, 1)
2 hosp2 <- rnorm(300, 6.5, 2)
3
4 # Test null hypothesis that variances are equal
5 var.test(hosp1, hosp2, alternative = "two.sided")
```

F test to compare two variances

```
data: hosp1 and hosp2
F = 0.22174, num df = 199, denom df = 299, p-value < 2.2e-
16
alternative hypothesis: true ratio of variances is not
equal to 1
95 percent confidence interval:
 0.1725330 0.2868607
sample estimates:
ratio of variances
 0.2217402
```

Testing Differences with Unknown, Unequal Variance

- What do we do if the variances are not equal?
- How would you interpret the results below?

```
1 # Use t-test with var.equal = FALSE to compare means
2 t.test(hosp1, hosp2, var.equal=F)
```

Welch Two Sample t-test

```
data: hosp1 and hosp2
t = 3.5032, df = 455.3, p-value = 0.0005052
alternative hypothesis: true difference in means is not
equal to 0
95 percent confidence interval:
 0.1993302 0.7087255
sample estimates:
mean of x mean of y
 7.078040  6.624013
```



Dependent/Paired Samples – Unknown Unequal Variance

Testing Differences for Dependent Samples

So far, we've discussed comparisons of independent samples, what about dependent or paired (e.g., pre/post) samples?

- Ex. You're testing the impact of a behavioral intervention on depression as measured on a validated scale ranging from 1-20. Participants are evaluated pre and post intervention.

```
1 set.seed(894894)
2 pre <- round(rnorm(30,14,3),0)
3 post <- round(rnorm(30,10,2),0)
4
5 t.test(pre,post, paired=T)
```

Paired t-test

```
data: pre and post
t = 7.6364, df = 29, p-value = 2.027e-08
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 3.099534 5.367132
sample estimates:
mean difference
 4.233333
```



Correlation

Correlation

- Often we are interested in whether 2 variables are linearly related.
- When x increases what happens to y ?
 - It increases.
 - It decreases.
 - Nothing
 - It varies with the value of x
- We can explore this by assessing correlation.
 - Correlation method used depends on assumptions.

Correlation methods and assumptions

- Pearson's correlation is one of the most commonly used correlation methods, but it has several important assumptions.
 - The data are continuous (i.e., not ordinal or categorical)
 - The data are linearly related
 - The data follow a bivariate normal distribution
- Spearman correlation is another commonly used method, but doesn't require the data to follow a bivariate normal distribution. It can also handle monotonic relationships and ordinal data.
- Kendall's tau is another non-parametric method (like Spearman), but is less commonly used and we will not cover it in this class.



Bivariate Normality

Bivariate normality and monotonic relationships

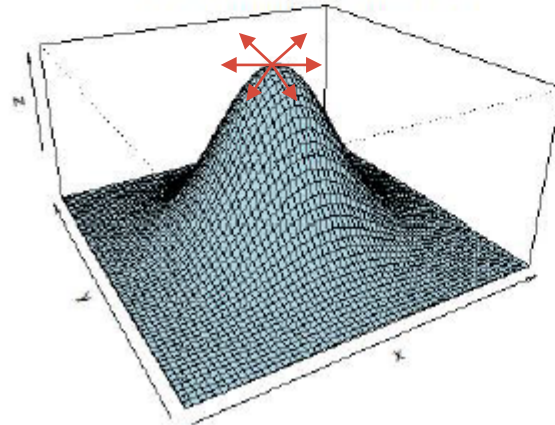
For bivariate normal random variables, x and y ,

$$\begin{pmatrix} x \\ y \end{pmatrix} \sim N \left(\begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}, \begin{pmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{pmatrix} \right) \Rightarrow \begin{matrix} x \sim N(\mu_x, \sigma_{xx}) \\ y \sim N(\mu_y, \sigma_{yy}) \end{matrix}$$

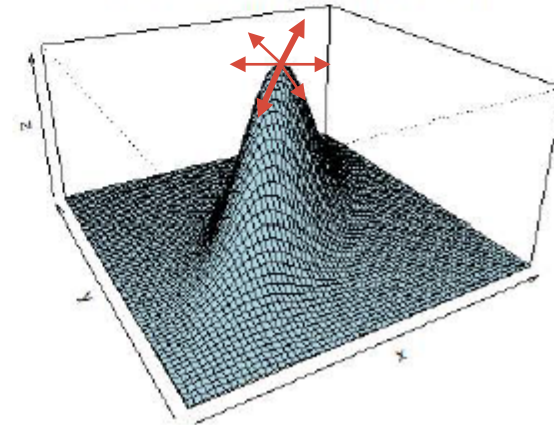
Often defined with the correlation coefficient, ρ_{xy} , ranging -1 to 1 rather than the covariance, σ_{xy} , as

$$\rho_{xy} = \frac{\sigma_{xy}}{\sqrt{\sigma_{xx}\sigma_{yy}}}, \text{ estimated with } r_{xy} = \frac{s_{xy}}{\sqrt{s_{xx}s_{yy}}}$$

Bivariate normal density, cor = 0

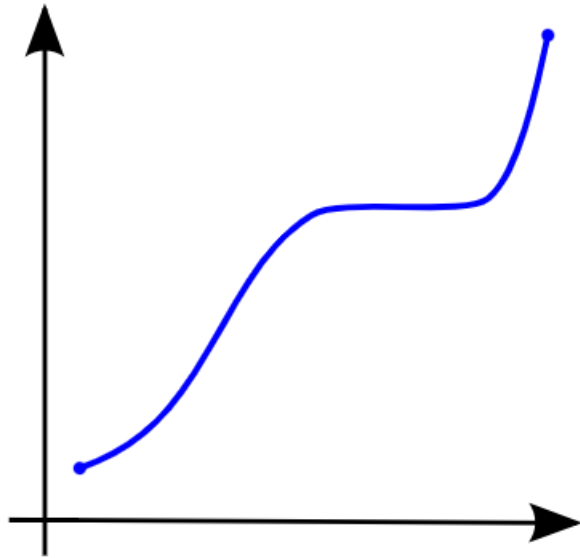


Bivariate normal density, cor = 0.75

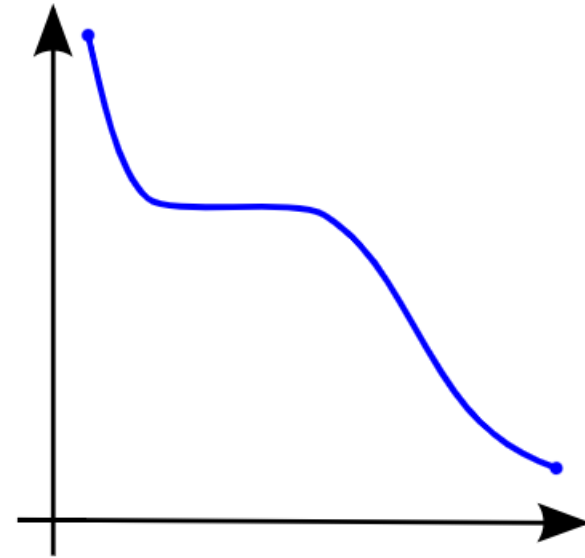


Monotonic relationships

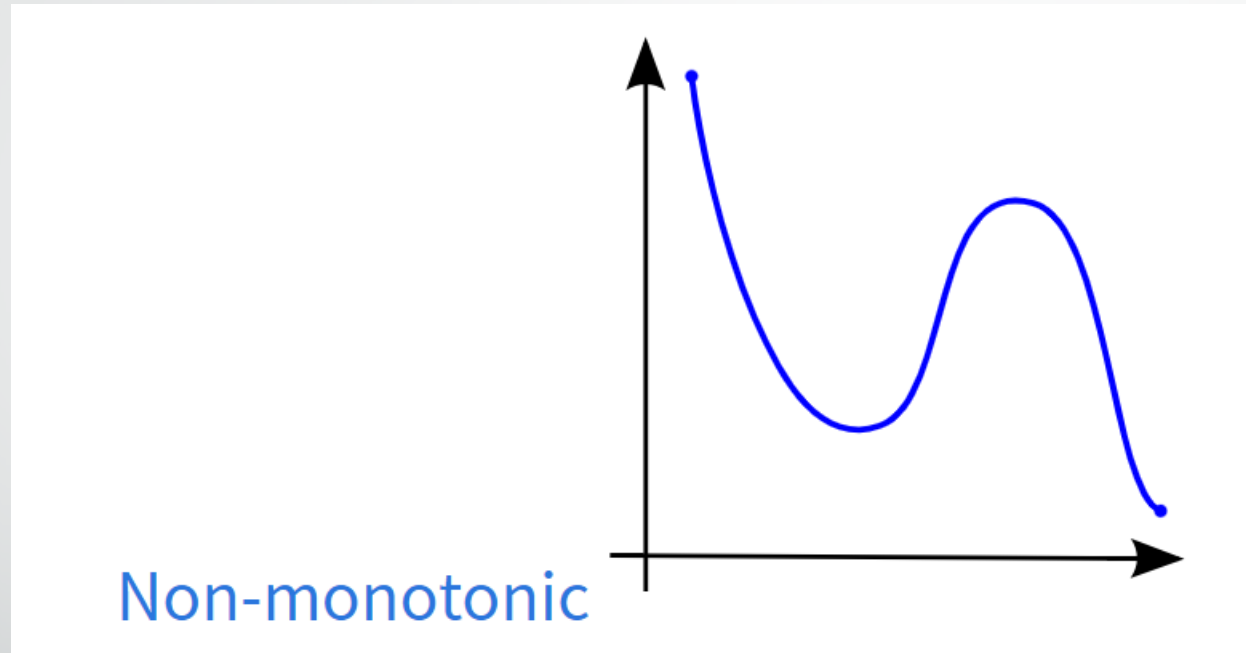
Monotonic increasing



Monotonic decreasing



Non-monotonic relationship



Hypothesis Testing with Correlation Coefficients

$H_0 : \rho_{xy} = \rho_0$ vs. $H_A : \rho_{xy} \neq \rho_0$ (Usually, $\rho_0 = 0$)

Use $r_{xy} = \frac{s_{xy}}{\sqrt{s_{xx}s_{yy}}}$ with $\hat{V}(r_{xy}) \approx \frac{(1 - r_{xy}^2)^2}{n - 3}$ under normality

Fisher's Z-statistic, for the case of $\rho_{xy} \neq 0$:

$$z_{xy} = \frac{1}{2} \ln \left[\frac{1 + r_{xy}}{1 - r_{xy}} \right] \approx N \left(\frac{1}{2} \ln \left(\frac{1 + \rho_{xy}}{1 - \rho_{xy}} \right), \frac{1}{n - 3} \right)$$

$$\rightarrow \text{Test-statistic} = \frac{Z_{xy} - z_0}{\frac{1}{\sqrt{n - 3}}}$$

$$\rightarrow 100(1 - \alpha)\% \text{CI: } \left[Z_{xy} - \frac{z_{\alpha/2}}{\sqrt{n - 3}}, Z_{xy} + \frac{z_{\alpha/2}}{\sqrt{n - 3}} \right]$$

Fisher's Z and Correlation Coefficients

$$Z_{xy} = \frac{1}{2} \ln \left[\frac{1 + r_{xy}}{1 - r_{xy}} \right]$$

$$\rightarrow r_{xy} = \frac{e^{2Z_{xy}} - 1}{e^{2Z_{xy}} + 1}$$

$$100(1 - \alpha)\% \text{CI of } Z_{xy}: \left[z_{xy} - \frac{z_{\alpha/2}}{\sqrt{n - 3}}, z_{xy} + \frac{z_{\alpha/2}}{\sqrt{n - 3}} \right]$$

$$\rightarrow 100(1 - \alpha)\% \text{CI of } \rho_{xy}: \left[\frac{e^{2z_{xy,Lower}} - 1}{e^{2z_{xy,Lower}} + 1}, \frac{e^{2z_{xy,Upper}} - 1}{e^{2z_{xy,Upper}} + 1} \right]$$



Test Statistics for Inferences

Test Statistics for Inferences

- When making inferences, are we interested in y or μ ?
- What do we need to consider?

After standardization (or studentization), the test statistic for $\bar{y}$ or any linear combination of $\bar{y}$, noted as $f(\bar{y})$, follows:

- $$\text{test stat} = \frac{f(\bar{y}) - \text{a value tested against in hypothesis}}{\text{variability in } f(\bar{y})}$$

E.g., $t = \frac{\bar{y} - \mu_0}{SE(\bar{y})} \sim N(0,1)$ under a large n

Test Statistic Examples

- 1 One sample with known variance: $Z = \frac{\bar{y} - \mu_0}{\sqrt{\sigma^2/n}}$
- 2 One sample with unknown variance: $t = \frac{\hat{\bar{y}} - \mu}{\sqrt{s^2 n^{-1}}}$
- 3 Two sample difference with unknown equal variance

$$t_{pooled} = \frac{\bar{x} - \bar{y} - (\mu_x - \mu_y)}{\sqrt{\left(\frac{1}{n_x} + \frac{1}{n_y}\right) s_{pooled}^2}}$$

- 4 Two sample difference with unknown unequal variance

$$t' = \frac{\bar{x} - \bar{y} - (\mu_x - \mu_y)}{\sqrt{\left(\frac{s_x^2}{n_x} + \frac{s_y^2}{n_y}\right)}}$$

- 5 Bivariate correlation coefficient

Use Z_{xy} , transformed r_{xy} : $Z_{xy} = \frac{Z_{xy} - z_0}{\frac{1}{\sqrt{n-3}}}$



Testing Normality

Testing Normality – 1

- Easier to assess univariate distribution which is what we will focus on.
- Should do so both statistically and graphically
- Two primary measures that we use:
 - Skewness: Departure from symmetry, $\frac{E(y-\mu)^3}{\sigma^3}$
 - Kurtosis: Peakedness of distribution, $\frac{E(y-\mu)^4}{\sigma^4}$
- Tested through:
 - Q-Q plots with Rankits
 - Shapiro-Wilk Statistic (W)
 - Tests H_0 : Random variable y follows a normal distribution
 - S-W test can be quite sensitive, particularly with large sample sizes
 - **Should always include visualization (Q-Q) in normality assessment**

$$W = \frac{\sum_{i=1}^n [(y_{i:n} - \bar{y})(m_{i:n} - \bar{m})]}{\sqrt{\sum_{i=1}^n (y_{i:n} - \bar{y})^2 \sum_{i=1}^n (m_{i:n} - \bar{m})^2}}; \text{ range: } [0,1]$$

Testing normality – 2

Shapiro-Wilk test of normality

- Tests H_0 : random variable follows a normal distribution
- Can be quite sensitive, particularly with large sample sizes.
- **Should always assess graphically too**

```
1 set.seed(894894)
2
3 shapiro.test(rnorm(50))
```

Shapiro-Wilk normality test

data: rnorm(50)
W = 0.97397, p-value = 0.333

```
1 shapiro.test(runif(50))
```

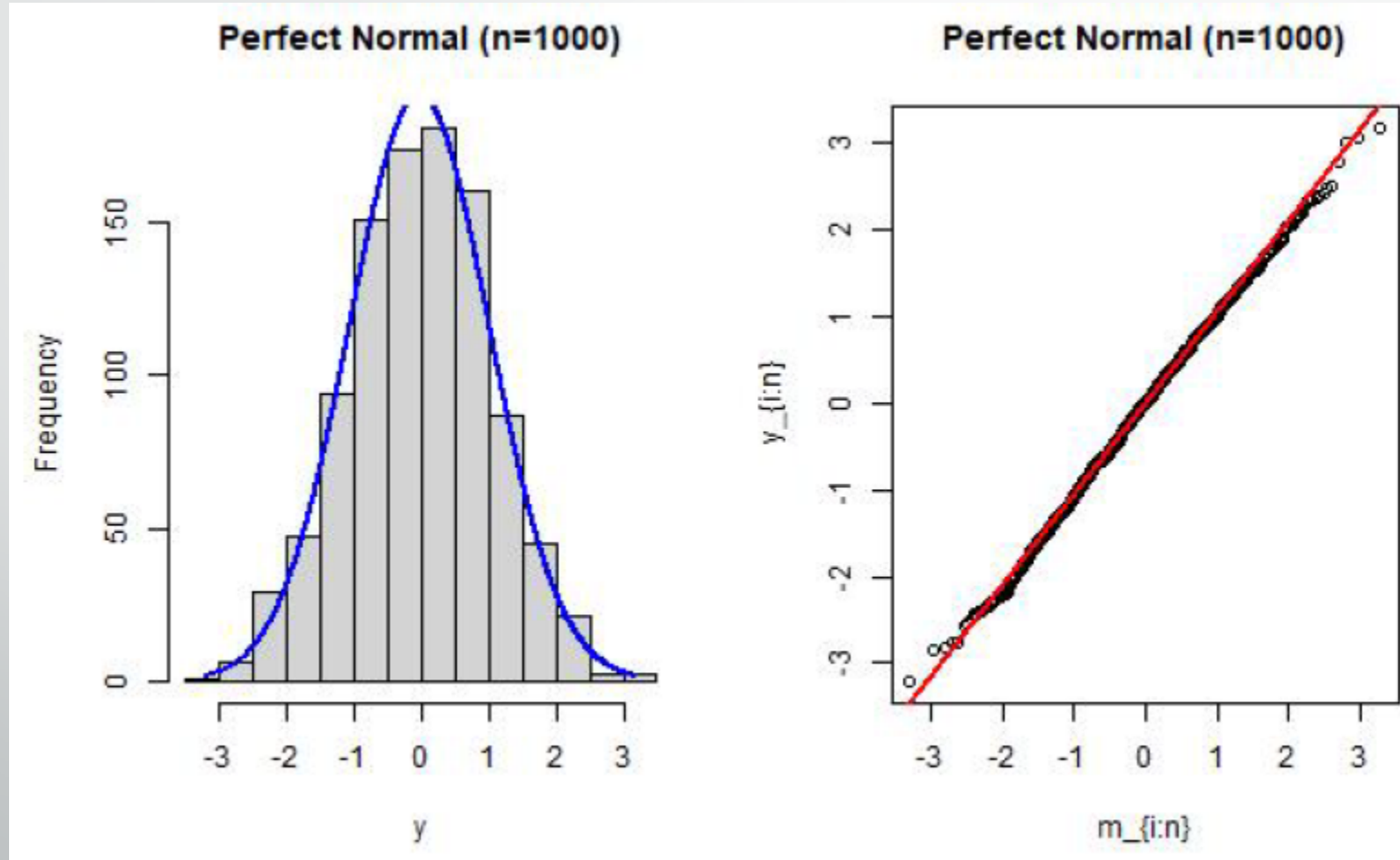
Shapiro-Wilk normality test

data: runif(50)
W = 0.95883, p-value = 0.07944

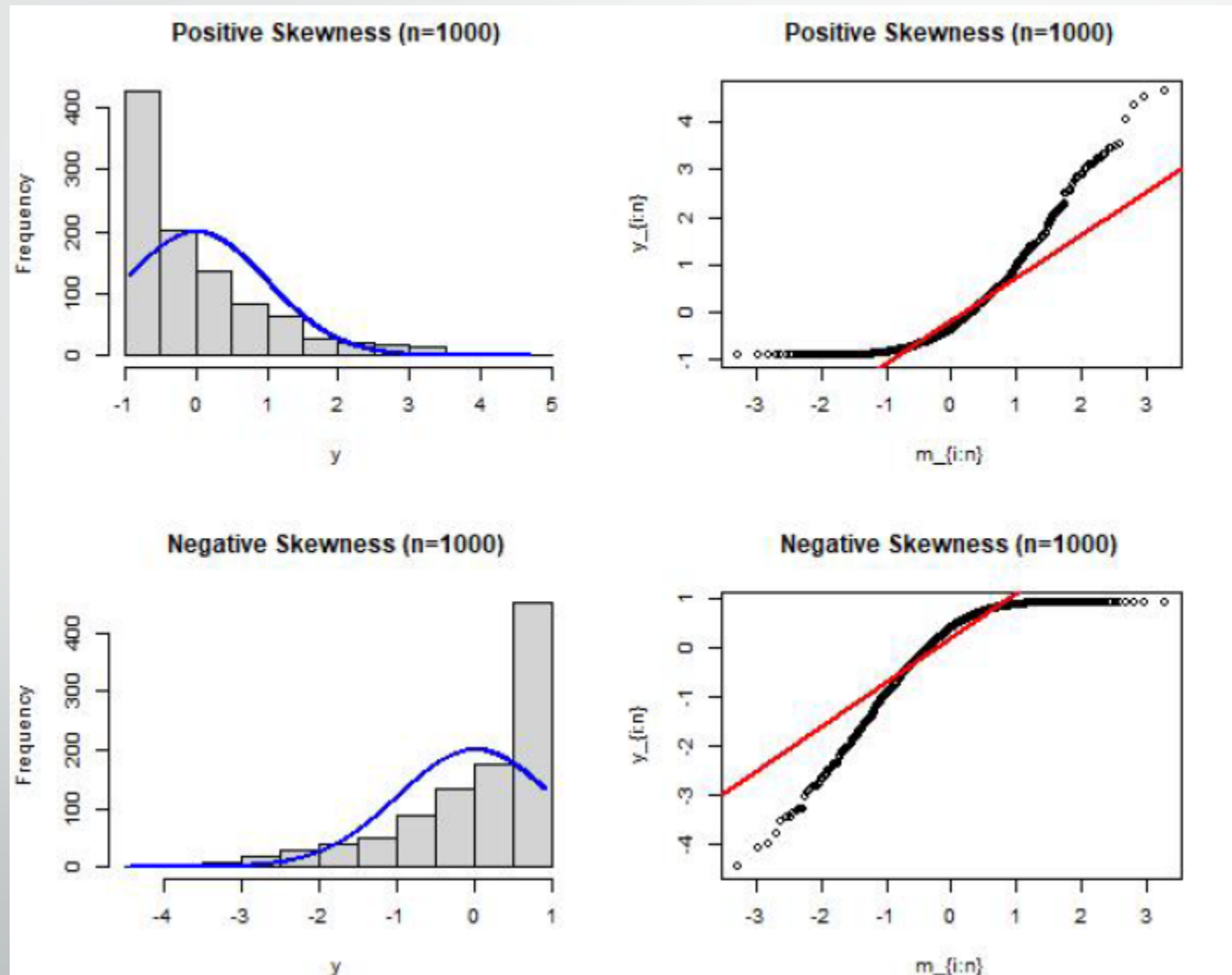
Assessing normality graphically

- Quantile-Quantile (Q-Q) plots are a method we use to compare distributions.
- Here we are using them to assess the normality of a distribution.
- They work by comparing the quantile for each y_i to the computed quantile of a hypothesized distribution (e.g., a normal distribution).
- Can then check whether the data follow this distribution.

Assessing normality graphically – Perfect Normal

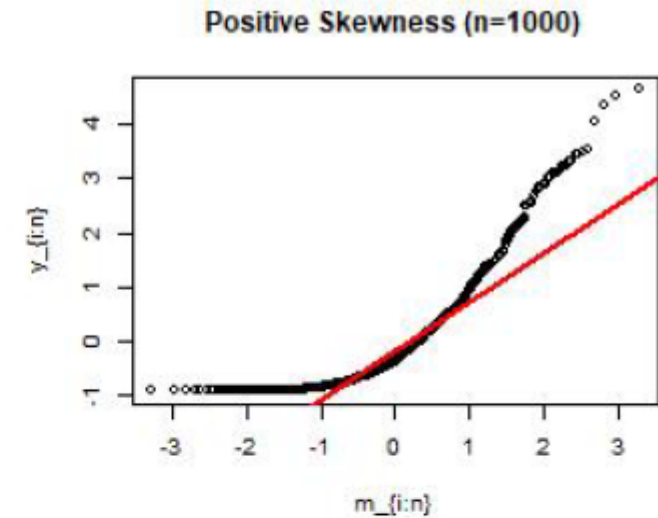
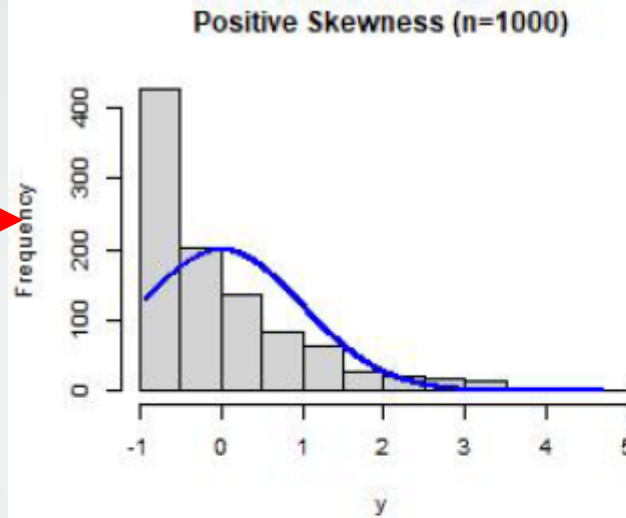


Assessing normality graphically – Skewed

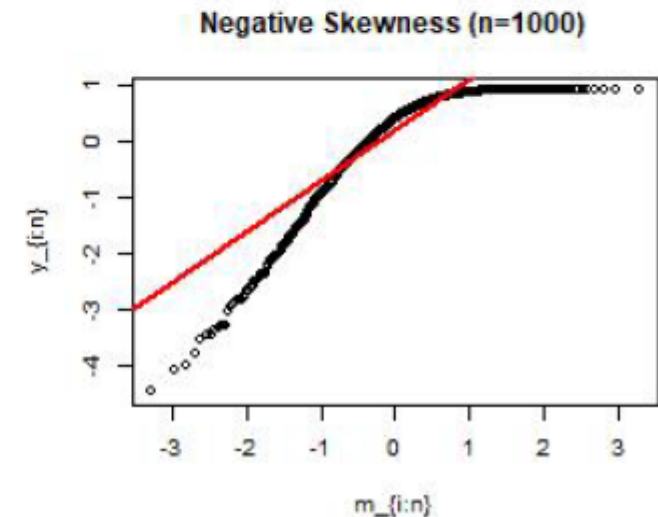
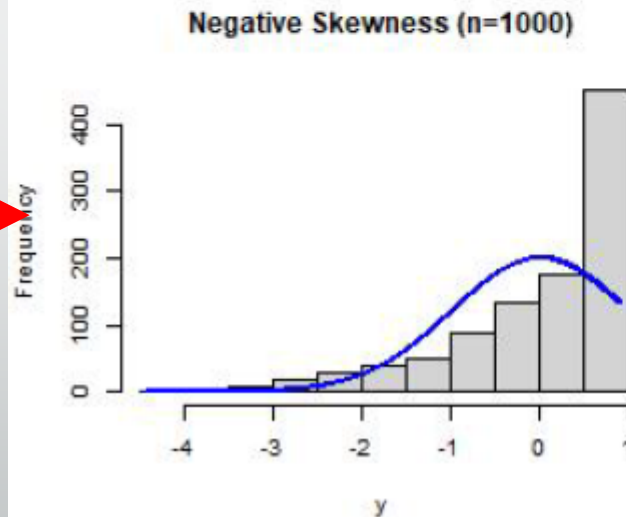


Q-Q/Rankit Plots – Skewness

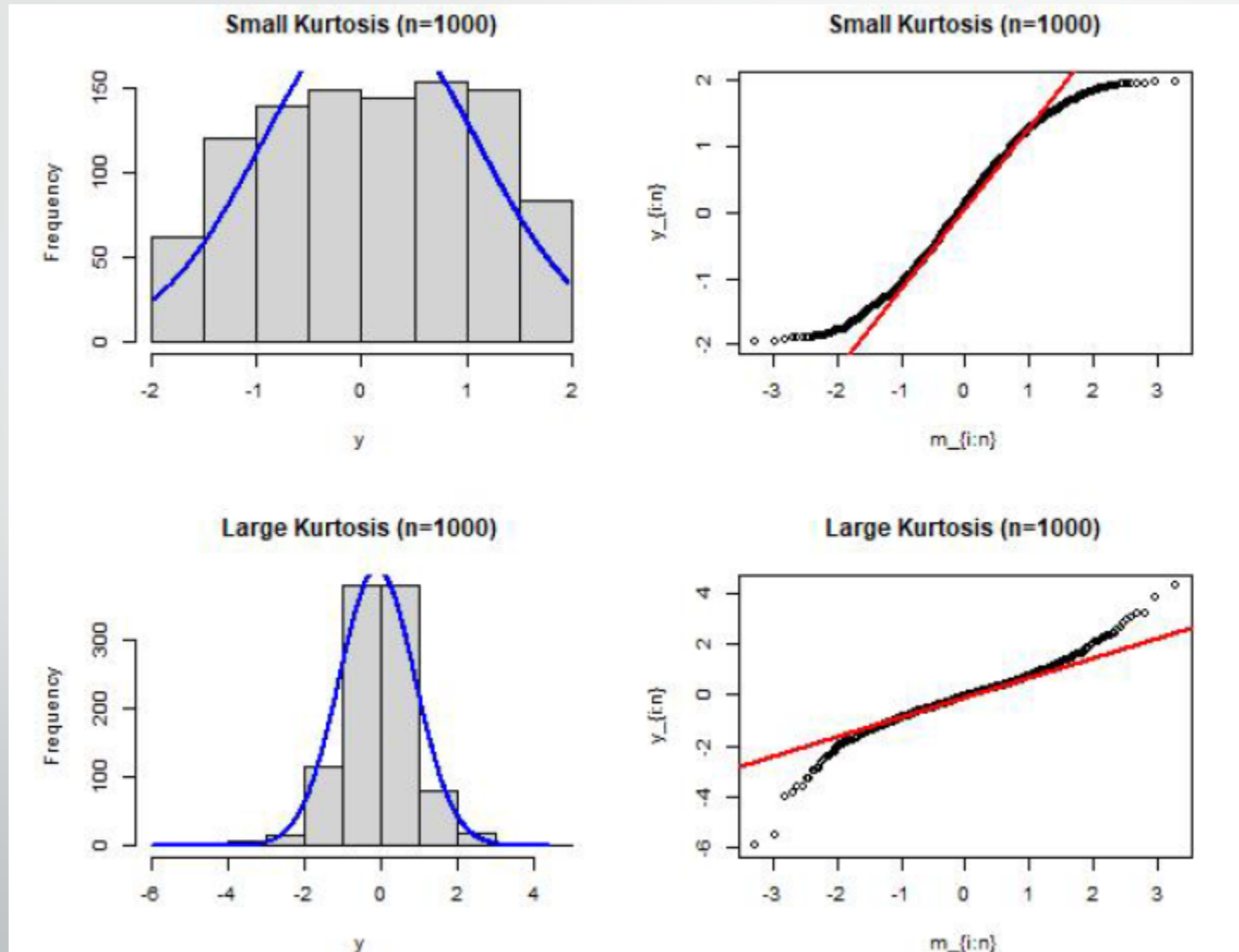
- AKA “right skewed”
- Distribution has long tail to the right
- $\text{mean} > \text{median}$



- AKA “left skewed”
- Distribution has long tail to the left
- $\text{mean} < \text{median}$

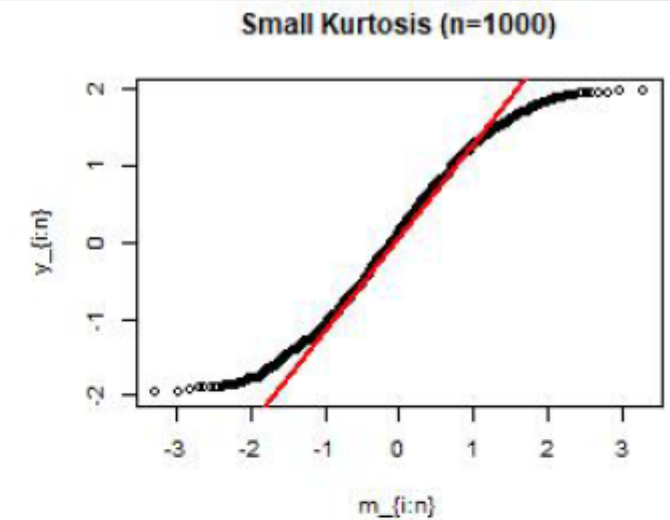
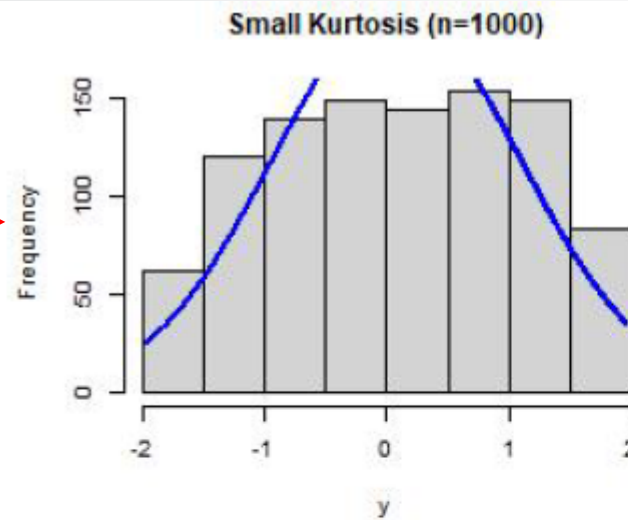


Assessing normality graphically – Kurtosis

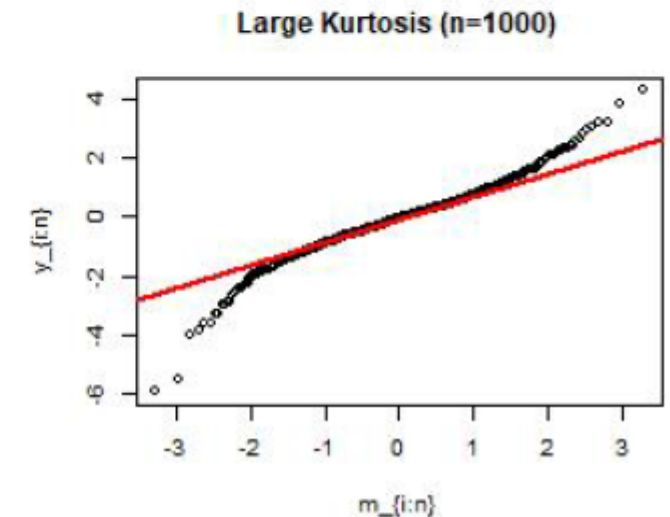
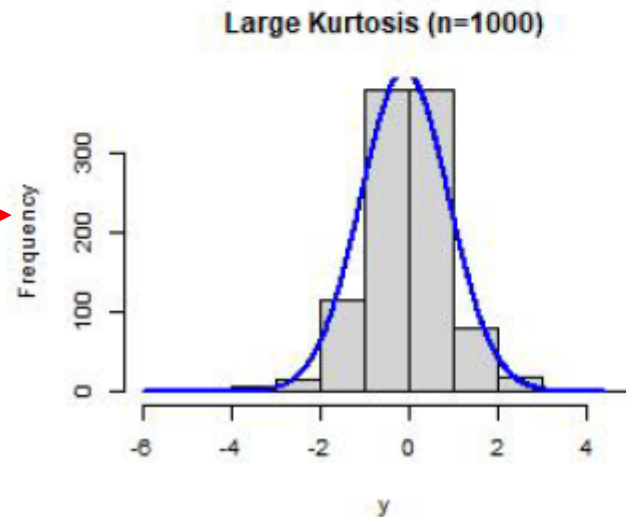


Q-Q/Rankit Plots – Kurtosis

Lighter tails, fewer and/or less extreme outliers



Heavier tails, many and/or more extreme outliers



More detail on Q-Q plots and their interpretation

- [Q-Q Plots and Worm Plots from Scratch](#)
 - Takes you through each step of making the plots
- [A Q-Q Plot Dissection Kit](#)
 - Q-Q plot examples and discussion of their interpretation



Why might departures from normality cause problems?



Lab session

How do these work when analyzing data?

- Let's turn to R: "Class2_Lab_nocode.Rmd"